NHS Connecting for Health	**Consultation Document – Suggestions for Implementing SNOMED CT into the User Interface of an Electronic Healthcare System**			
	Programme	Data Standards and Products	**Document Record ID Key** **NPFIT-FNT-TO-DSD-0218.01**	
	Sub Programme	SNOMED CT Deployment		
	Project Manager	Steve Bentley	**Status**	First version
	Owner	Steve Bentley	**Version**	1.00
	Authors	Martin Rooney	**Version Date**	04 November 2011

Consultation Document

Suggestions for Implementing SNOMED CT into the User Interface of an Electronic Healthcare System

Common User Interface team
NHS Connecting for Health

© Crown Copyright 2011

Amendment History:

Version	Date	Amendment History
1.00	04-11-2011	First version

Reviewers:

This document must be reviewed by the following:

Name	Signature	Title / Responsibility	Date	Version
Steve Bentley		Head of BDD, Data Standards and Products	04-11-2011	1.00

Approvals:

This document must be approved by the following:

Name	Signature	Title / Responsibility	Date	Version
Steve Bentley		Head of BDD, Data Standards and Products	04-11-2011	1.00

Document Control:

The controlled copy of this document is maintained in the NHS CFH Electronic Document and Records Management System. Any copies of this document held outside of that system, in whatever format (e.g. paper, email attachment), are considered to have passed out of control and should be checked for currency and validity.

Glossary of Terms:

Terms, acronyms and abbreviations commonly used within NHS CFH can be found in the Glossary of Terms http://intranet.connectingforhealth.nhs.uk/departments/npo/glossary/glossary. If any terms are used in this document that are not currently included in the glossary please send them to cfh.glossary@nhs.net.

© Crown Copyright 2011

Contents

1 Introduction....4
1.1 Purpose....4
1.2 Relationship to Existing CUI Guidance....4
1.3 Background....4
1.4 Intended Audience....5
1.5 Providing Feedback....5
1.6 Source of the Suggestions....5
1.7 Scope....5
1.7.1 In-Scope 5
1.7.2 Out of Scope 6
1.8 Assumptions....7
1.9 SNOMED CT Encoding....7
2 Suggestion Details....9
2.1 Conformance Indicator....10
2.2 Evidence Rating....10
3 Suggestions....11
3.1 SNOMED CT Data and Context....11
3.2 Entering a Search Term....14
3.3 Matching Concepts....21
3.4 Displaying Search Results....26
3.5 Refining the Search....38
3.6 Selecting a Result....40
4 Points for Further Discussion....42
5 References....43

© Crown Copyright 2011

1 Introduction

1.1 Purpose

This document presents a list of draft suggestions for how SNOMED CT encoding can be implemented successfully into an electronic healthcare system (EHS). The suggestions focus on improving the usability and safety of systems providing SNOMED CT single-concept matching encoding functionality. They cover encoding systems that use SNOMED CT terms in a pre-coordinated format. Post-coordinated encoding is out of scope for this work.

This document is the first step in a project that through clinical consultation and research will aim to define a set of implementation guidelines describing how best to include SNOMED CT encoding in electronic healthcare systems.

The suggestions provided are not final recommendations, but merely a starting point for debate. It is hoped that by sharing views and opinions based upon the contents of this document that a more refined set of suggestions can be produced. To this end, this document also describes how feedback can be submitted to the authoring team.

1.2 Relationship to Existing CUI Guidance

Several NHS Connecting for Health Common User Interface (CUI) documents already exist providing guidance on the searching, matching and displaying of clinical codes in a user interface.

This document does not replace those existing CUI guidance documents. This document focuses solely on single-concept encoding and should be seen as an accompaniment to the existing ones.

Whilst there is some overlap in the topics presented in this document and existing CUI guidance documents, the suggestions given in this document do not yet contain a sufficient level of detail to allow implementation by a software vendor wanting to include SNOMED CT into the design of their healthcare systems.

1.3 Background

The term “encoding system” is used throughout this document. For the purpose of this work, this should be taken to mean an electronic healthcare system that allows the clinician to record their clinical findings using a clinical reference terminology (like SNOMED CT).

Many NHS healthcare systems are encoding systems – that is they already use some form of terminology coding so that clinicians can record precise clinical information in a patient's electronic health record. The success of these systems will depend on several factors, one of those being its usability.

If the coding process interferes too much with the clinician’s primary task at hand (most often their interaction with their patient), or it takes so much time as to delay

© Crown Copyright 2011

their already busy workflow, or it simply cannot perform as expected, then the system will be rejected by clinicians.

This document – and the wider piece of work – aims to produce a set of guidelines that, when implemented into the user interface design of healthcare systems, will improve its overall usability and safety.

1.4 Intended Audience

The intended audience for this document is anyone interested in the implementation of clinical terminology encoding in healthcare systems.

1.5 Providing Feedback

As stated, this document is a first step to defining a set of SNOMED CT implementation guidelines. By sharing views and opinions, the suggestions in this document can be matured in order to progress them to the next project phase.

If you have any comments or feedback on any part of this document, then please send them to the email address below. All comments will be kept confidential.

Send to: cuistakeholder.mailbox@nhs.net

1.6 Source of the Suggestions

The following sources were used to create the list of draft suggestions.

- Discussions with UK Terminology Centre SNOMED CT subject matter experts.
- Short usability reviews of healthcare systems that already implement encoding using a clinical reference terminology.
- Reference to clinical reference terminology implementation literature (see References section of this document).
- Existing NHS Connecting for Health guidance and standards in the area of encoding (see References section of this document).

1.7 Scope

1.7.1 In-Scope

The following list forms the scope of this work.

- **Electronic Healthcare record** - use of SNOMED CT within the context of an electronic healthcare system record.
- **Point of care encoding** - the suggestions cover encoding at the point of care where the clinician records the patient consultation.

© Crown Copyright 2011

- **Single-concept-matching** - the suggestions are based upon 'single-concept-matching' where the clinician searches a single concept (e.g. 'chronic obstructive lung disease').
- **Pre-coordinated terms** - the suggestions assume that the encoding system uses SNOMED CT terms that are in a pre-coordinated format.
- **Primary and Secondary Care** - whilst it is the case that many NHS Primary Care systems already use some form of terminology coding, they do not all work in the same way and none of them currently use SNOMED CT in their implementation. Because of this, it is assumed that the suggestions provided in this document are applicable for both Primary and Secondary Care.
- **Searching via text entry** - the clinician will search for the wanted concept using a text entry input mechanism (i.e. where they type in some characters that form the search input).
- **Browsing** – whilst searching via text entry is the main method described in this document for clinicians to locate the concept they want, these suggestions also cover the use of browsing to locate a concept.
- **Positive and Negative Suggestions** - whilst the majority of the suggestions presented give guidance on what should be done in order to implement SNOMED CT in the user interface of an EHS, some suggestions give guidance on what should not be done. These 'negative' suggestions then include specific practices that are not advised or that should be deprecated.
- **Usability & safety** - the suggestions focus on the usability and safety aspects of implementing SNOMED CT encoding functionality.

1.7.2 Out of Scope

The following are not within the scope of this work.

- **Browsers** – implementing SNOMED CT into a standalone browser.
- **Coding at a later moment** – such as done by a dedicated 'coder'.
- **Free-text parsing** – where the clinician can type in unconstrained textual notes that the system matches to SNOMED CT expressions.
- **Post-coordinated terms** - the suggestions do not cover post-coordination of SNOMED CT terms.
- **Non-Text entry search** - the suggestions do not cover other possible, non-text entry search mechanisms, such as selecting SNOMED CT terms directly from a pre-defined pick-list.
- **Technical implementation** - the suggestions do not cover the technical side of implementing either SNOMED CT or encoding using SNOMED CT into an electronic healthcare system. Specific details of search algorithms are beyond the scope of this document.
- **Reporting No Matches Found** – this work does not include any reporting mechanisms to those who manage and maintain SNOMED CT for when a search does not return a result.

© Crown Copyright 2011

1.8 Assumptions

The suggestions presented are subject to the following assumptions.

- **Large number of SNOMED CT terms** – it is assumed that the search process occurs across a large number of SNOMED CT terms (e.g. 2000 terms or more).

  Several specialties, such as A&E, Dermatology, Endocrinology and Infections will typically have terminology subsets of 2000 or more terms.

  If only a small number of terms are to be searched across, then the text entry search mechanism may not be the most efficient way for encoding terms.

- **Other Clinical Terminologies** – whilst this work focuses on the implementation of SNOMED CT in the user interface of healthcare systems, it is thought that many of the suggestions presented are equally applicable to other clinical terminologies (e.g. READ).

- **User with Limited Coding Experience** – it is assumed that we are designing for a novice clinical user with limited knowledge and expertise related to clinical coding and SNOMED CT.

1.9 SNOMED CT Encoding

For this document, coding is taken to cover the processes of searching, matching, displaying, browsing, refining and selecting SNOMED CT concepts. This will typically involve a clinician entering a few characters of the term they want to find, triggering the search, browsing the matched results and then selecting one of the results.

In general, the characteristics that make up an 'ideal' encoding experience will include some or all of the following:

- The clinician understands the encoding interface – they intuitively know how and where to search.
- Just by typing the first few letters of each search word a manageable number of results is produced.
- The number of results should be small enough for the user to review quickly and to find things easily.
- The results list should include the term that the clinician is seeking to record (provided that a match exists within the terminology database).
- The most likely results should be towards, or at the top of, the results list.
- The ordering or grouping of the results list should reflect the user's expectations and needs.
- The results list should have high specificity (i.e. present all the expected results). It should also have a low false-positive rate (i.e. few, if any results which appear irrelevant).
- The time taken to generate and display the results list must be fit for the intended purpose, so in a clinical setting should be say less than 3 seconds.

© Crown Copyright 2011

- Each term displayed in the results list contains enough detail to display it clearly and unambiguously. It could not be mistaken for another, similar term.

The reality is that no actual encoding system currently implements a clinical terminology – particularly of SNOMED CT – in such an ideal way. Systems will often leave clinicians wondering how they need to perform the search. Searches can be slow and unresponsive leaving the clinician wondering if anything is happening. Result tables can list hundreds of results, often in a random order. This document attempts to address these types of usability problems and to produce a list of suggestions that strive towards the ideal characteristics

© Crown Copyright 2011

2 Suggestion Details

Below outlines some details that should be noted when considering the suggestions.

- Most of the suggestions are not fully complete with enough detail that would allow them to be implemented by a healthcare system software vendor.
- Several suggestions are a purely a statement of a requirement, they do not always provide a pragmatic solution on how that requirement should be solved or implemented in a system.
- Some suggestions include a diagram to aid understanding. The visual representation of any suggestion diagrams is for illustrative purposes only. Stylistic choices (e.g. font and colours used) should not be taken as part of the suggestion
- Some of the suggestion diagrams appear with either a tick or a cross next to them. The significance of these is as follows:



A correct usage example



An incorrect usage example

- The suggestions have not been subjected to a formal patient safety assessment. Such an assessment is expected at a later stage in the project.
- Many suggestions could be considered as simply being ‘common sense’, with almost no need to document them. Whilst, this may appear the case, it is surprising how many existing encoding systems do not follow these common sense rules.
- Each suggestion is presented with a title bar that contains an identification code, a short descriptive title, a conformance indicator and an evidence rating indicator (see Figure 1). The sections below describe the conformance and evidence rating indicators and the values that they can have.

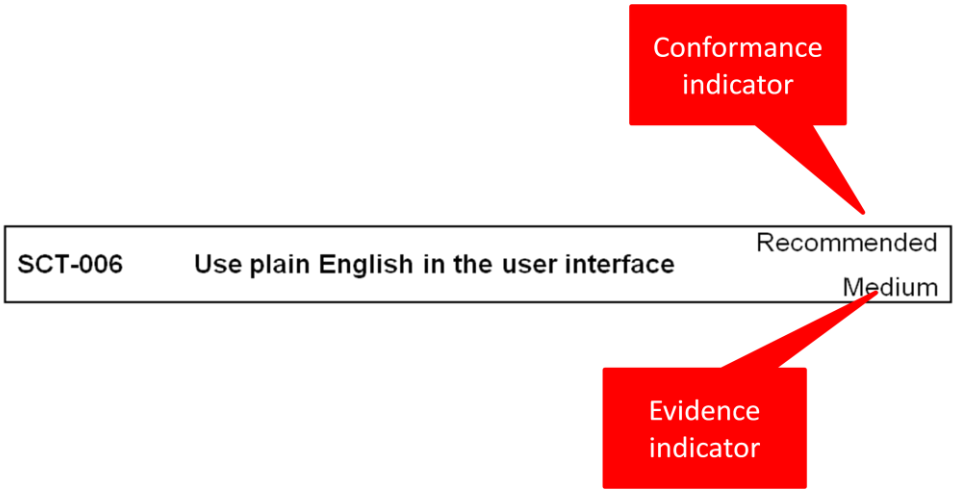


Figure 1: example of a suggestion title bar.

© Crown Copyright 2011

2.1 Conformance Indicator

Conformance indicates the extent to which a software vendor should follow the guideline when implementing SNOMED CT encoding in the user interface of their electronic healthcare system. There are 2 conformance levels:

Mandatory – an implementation should follow the suggestion.

Recommended – whilst an implementation is still advised to follow the suggestion, it is optional as to whether they do so or not.

2.2 Evidence Rating

The Evidence rating summarises the strength of the research defining the suggestion and the extent to which it mitigates safety hazards. The values this can take are listed below (with example factors used to determine the appropriate rating):

High

- Strong agreement between the majority of SNOMED CT subject matter experts.
- Early existing system reviews gives strong indication that suggestion would improve design usability or mitigate specific patient safety hazards.
- Referenced usability principles indicate the suggested design is significantly better than alternatives.

Medium

- Some agreement between SNOMED CT subject matter experts.
- Some indication from early existing system reviews that suggestion would improve design usability or mitigate specific patient safety hazards.
- References some recent authoritative guidance or research (e.g. CUI).
- Referenced usability principles indicate the design is significantly better most alternatives.

Low

- Low or no real agreement between SNOMED CT subject matter experts.
- Low or no indication from early existing system reviews that suggestion would improve design usability or mitigate specific patient safety hazards.
- Referenced usability principles indicate the design is not significantly better than alternatives.

Idea

- No evidence that suggestion would improve usability or safety issues. Simply a proposed idea for reader feedback and further exploration.

Option

- More than one possible option for a suggestion is available.
- No certainty as to which one (if any) should be implemented.

© Crown Copyright 2011

3 Suggestions

This section presents the list of suggestions for implementing SNOMED CT in a usable way into an electronic healthcare system.

3.1 SNOMED CT Data and Context

The following are a list of suggestions surrounding the data used and its context.

SCT-001	**Use the current version of SNOMED CT**	Mandatory High

The UKTC very strongly advises that the current release of SNOMED CT should be used at all times.

It should not be the responsibility of the clinical users of SNOMED CT systems to ensure that an up-to-date SNOMED CT version is being used. If the version of SNOMED CT is out of date or incorrect, the user does not need to be notified or made aware of this. This information should be passed on to the appropriate system administrator.

The UK Edition of SNOMED CT should be used – which comprises the UK Clinical Extension plus the UK Drug Extension, and the International Core.

Rationale: the current version of SNOMED CT should always be used to prevent any interoperability problems arising from systems using different versions. Also, using the latest version ensures that clinicians can benefit from new concept additions.

It is also part of the UKTC SNOMED CT licensing agreement that upon receipt of a notification that an update, or within 180 days that a new version is available, the licensee shall promptly incorporate the SNOMED CT update or new version into their products.

SCT-002	**Do not require nor allow the user to choose the clinical terminology version**	Mandatory Medium

Do not ask the user to decide which clinical terminology system or version to use. The clinical terminology should be automatically set – the user should never be asked or offered to choose a different one.



Figure 2: do not allow the user to choose the clinical terminology to be used as in the example above.

Rationale: the users should not have to concern themselves about whether the correct clinical terminology is being used. Asking a user to choose a terminology just adds unnecessarily to their workload.

© Crown Copyright 2011

SCT-003 Exclude inappropriate content	Mandatory High

Certain terms should never be browseable or selectable in any clinical context as the primary code to record a clinical finding or procedure. Such codes are almost only ever clinically plausible in a post-coordinated context, where they may be used to elaborate upon some previously selected primary code.

Examples of SNOMED content that should never be browseable or selectable (except in a post-coordinating context) include:

- Non human content (e.g. Hump of Camel)
- Navigational concepts (e.g. Poisoning / Injury)
- Organisms (e.g. Salmonella)
- SNOMED Attributes (e.g. Cuff deflated)
- Record artefacts (e.g. Kick chart)
- Physical forces (e.g. High temperature)
- Anatomical structures (e.g. Osteophyte)
- Action qualifiers (e.g. Cryosurgery)
- Staging scales (e.g. Vital organ failure)
- UKTC former limited status subset

Suppressing these top level hierarchies entirely from appearing for selection - including when no user configurable contextual filters are applied - is very strongly recommended.

Rationale: the task of choosing a concept from the results list should not be made more complex and time consuming for the user by including items that should not be selected as a primary code.

SCT-004 Exclude inactive concepts	Mandatory High

Those SNOMED CT concepts that are inactive should not be browseable or selectable for the user.

Rationale: the task of choosing a concept from the results list should not be made more complex and time consuming for the user by including items that should not be selectable.

SCT-005 Present only SNOMED CT concepts relevant to context	Mandatory High

The default option should be that the coding process only includes those SNOMED CT concepts that are relevant to the clinician's current context of work. Avoiding many irrelevant concepts will help reduce the workload on the clinician and make it easier to find the desired term.

The context setting should be done automatically by the software system (e.g. recognising the logged on user and only presenting SNOMED CT terms related to their specialty). Context setting can be set on a variety of factors, such as clinical

© Crown Copyright 2011

specialities, healthcare provider (e.g. consultant or GP), the patient and corporate requirements where different clinical practices may have different requirements for encoding.

Rationale: the task of choosing a concept from the results list should not be made more complex and time consuming for the user by including items that do not belong in the current context.

SCT-006 Use plain English in the user interface	Recommended Medium

Do not use SNOMED CT language or jargon in the user interface as most users will not be familiar with this and won't necessarily understand it. The following are examples of SNOMED CT words or phrases that most users will not understand:

• Terminology axis	• Fully-specified-name
• Concept	• Preferred term
• Concept-Id	• Synonym
• Qualifier	

Rationale: to ensure that all text content is readable and understandable.

SCT-007 Use only English language SNOMED CT concepts	Mandatory Medium

The results list should only display **en** and **en-GB** SNOMED CT terms. An exception to this is to allow the displayed in en-US (American terms) when an American term is available and a UK one is not.

Descriptions available	**Display**
en	en
en, en-gb	en-gb
en, en-gb, en-us	en-gb
en, en-us	en
en-us	en-us

Figure 3: table showing conditions under which different language versions are to be displayed.

Rationale: to avoid any confusion and any ambiguity, a single, common encoding language should be used in clinical noting systems within the NHS.

© Crown Copyright 2011

3.2 Entering a Search Term

This section presents suggestions relating to the entering of a search term. As stated, this only covers searching for a concept by means of text entry.



Figure 4: text entry method of searching for a term by typing in characters for one or more words.

SCT-008	Ensure it is clear which text entry fields are encodable	Recommended Idea

Clearly distinguish in the user interface between those fields which are SNOMED CT encodable and those which are not (e.g. normal text entry fields).

Whilst encodable text fields and normal text fields essentially look the same, they do require different actions from users. In an encodable field, the user need only type in a few characters of the concept they want before triggering the search. They may also use shortcuts (e.g. abbreviations) to locate the wanted concept. In a normal text field users will have to type out in full what they want to say.

If users recognise in advance that a field is encodable, they can modify their typing behaviour accordingly.

Rationale: so that a user is aware in advance of how they can best enter their text and how the field will respond to it.

SCT-009	Ensure easy search triggering	Recommended Low

As well as displaying a button next to the text entry field, ensure that the search can be triggered by the user pressing the <enter> key.

Rationale: good keyboard implementation can improve user productivity. After typing in a search string it is very efficient if the user can simply hit the <enter> key rather than having to return to the mouse and click a button.

SCT-010	Ensure that the text entry field can hold sufficient characters	Mandatory Medium

The text entry field should be able to display 32 characters and hold up to 255 characters in total. If the user types beyond the 32 character visible limit, the field scrolls from left-to-right (but without a scroll bar), in order to ensure that the last character that the user has typed is visible.

Rationale: by ensuring that 32 characters are visible, this means that the full text of about 50% of SNOMED CT labels (Preferred term and Synonyms of active terms) can be displayed. By ensuring that the text entry field can hold a maximum of 255 characters, this means that the full length of all SNOMED CT labels can be accommodated.

© Crown Copyright 2011

SCT-011	Two character minimum	Recommended Low

The search should only be triggered after the user has entered a minimum of 2 characters in the search term. These characters should not be whitespace or blank characters.

Rationale: this limit prevents the user searching after just typing in 1 character which would return a very large amount of results. The 2 character limit is still low enough to still allow the user to enter 2 character abbreviations (e.g. **mi** for myocardial infarction).

SCT-012	Allow the entry of single or multiple search tokens	Mandatory High

The user should be able to enter either single (e.g. **diab**) or multiple (e.g. **com acq pneu**) search tokens.

Rationale: the entry field should automatically be able to handle however many search tokens the user decides to enter as a search term.

SCT-013	Search token order independent	Mandatory High

The user should be able to enter search tokens in any order (e.g. **fract ankle** and **ankle fract** should both be allowed and return the same results).

Rationale: requiring the user to stop and think about which order to enter search tokens in increases their mental workload and slows them down. This is an example of where the system can easily perform this task and search for all token order variations.

SCT-014	Do not require the user to enter wildcard characters to enable partial matching	Mandatory High

The user should not have to add any wildcard characters or symbols to the search string to denote partial matching. Partial matching should be enabled by default.

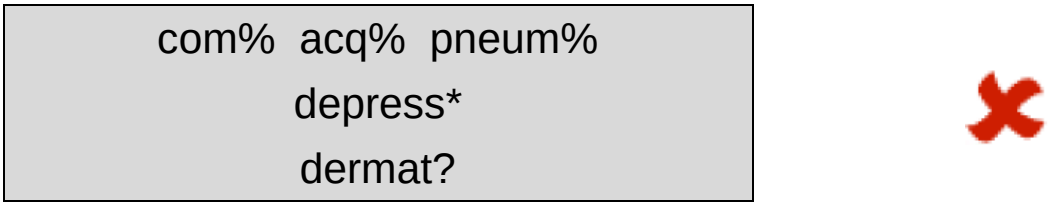


Figure 5: examples of search text entered including wildcard characters. The user should not have to use these characters at all.

Rationale: it should not be necessary for users to enter wildcard characters to be able to perform partial term matching. Most users will not be familiar with these characters. Asking a user to add such a character is also giving the user another, unnecessary task to perform.

© Crown Copyright 2011

SCT-015 Offer case-insensitivity	Recommended Medium

The system should not require the user to have to apply capitals or lower-case letters to search terms.

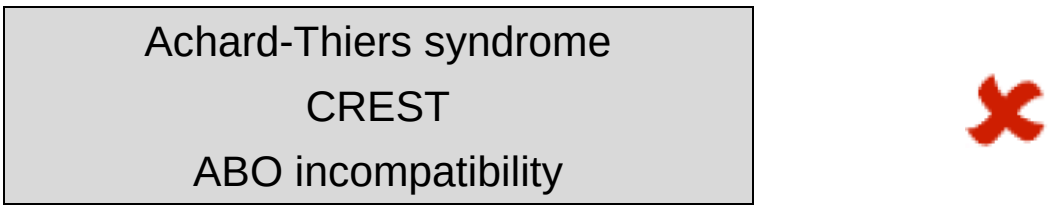


Figure 6: examples of search text that has been entered with case-sensitive letters. Users should not have to do this.

Rationale: the entry field should automatically be able to handle whatever case-characters the user chooses to enter. Forcing the user to enter in a specific case in order to retrieve specific matches is giving the user unnecessary, additional workload.

SCT-016 Prevent searching by SNOMED CT code	Mandatory Medium

Do not provide the ability for users to search by SNOMED CT code. Most users will not work with, nor remember SNOMED CT codes as they are random / non-related numbers.

Rationale: whilst there may appear no real harm in letting a user search by code, there exists the risk that users may easily mistype a SNOMED CT code and thus add an incorrect concept into the patient record.

SCT-017 Copy and Paste	Recommended Medium

Allow the user to copy and then paste text into the search field.

Rationale: this will enable easier and quicker searching for terms without the need for typing where users are referring to other electronic sources (e.g. copying from another system or a document).

SCT-018 Handle entered foreign characters	Recommended Medium

The user should be able to type in or copy and paste foreign characters as part of their search text. For example, they should be able to enter **Sjög** in order to locate the SNOMED CT concept **Sjögren-Larsson syndrome (disorder)**.

Rationale: for situations where the user types in search text (or copy & pastes it from another document) containing foreign characters the system should be able to

© Crown Copyright 2011

handle these without requiring the user to replace the foreign characters with English ones.

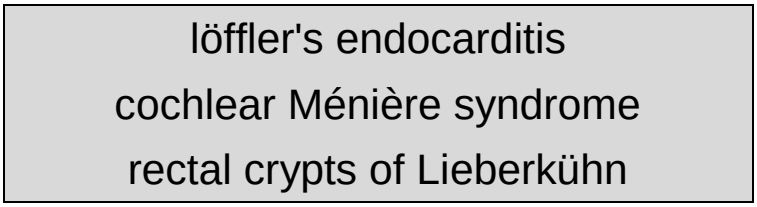


Figure 7: examples of search text that has been entered including foreign characters. Users should be able to do this.

SCT-019 Allow replacing of foreign characters with English ones	Mandatory Medium

Users should also be able to search *without* having to enter foreign characters where a SNOMED CT description contains them.

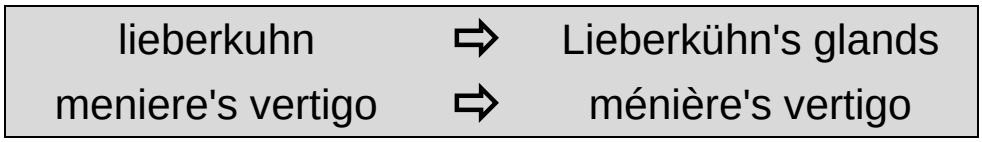


Figure 8: examples of search text that has been entered by the user with foreign characters replaced by English ones (left hand side). The SNOMED CT concepts (right hand side) that should be found.

Rationale: this is necessary as not all SNOMED CT concepts with a foreign character synonym have an equivalent synonym without the foreign characters.

SCT-020 Handle entered superscript and subscript characters	Recommended Low

The system should be able to handle the situation where the user types in or copy & pastes superscript or subscript characters in their search term.

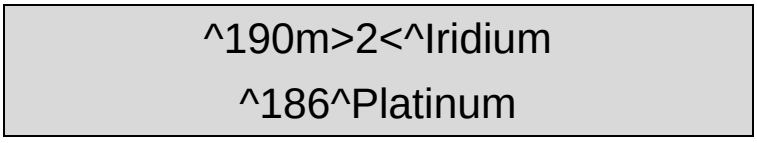


Figure 9: examples of SNOMED CT terms containing superscript and subscript characters.

It should be noted that this suggestion is not stating that the search must use these characters when matching. For example, depending on how the SNOMED CT db is tokenised, it may make more sense to strip them out and not use them. This suggestion is simply stating that the user be able to enter them, and that the system recognises them when entered and understand how to process them.

Rationale: certain departments (e.g. nuclear medicine or radiology departments) will require the ability to enter superscript (^ ^) and subscript (> <) characters.

© Crown Copyright 2011

SCT-021	Handle entered non-alphanumeric characters	Recommended Medium

The user should allowed to enter non-alphanumeric characters (e.g. those outside then range of 0-9, A-Z, a-z) in the search field.

The non-alphanumeric characters that can be entered could come from an approved list which would make implementing this suggestion easier for software vendors.

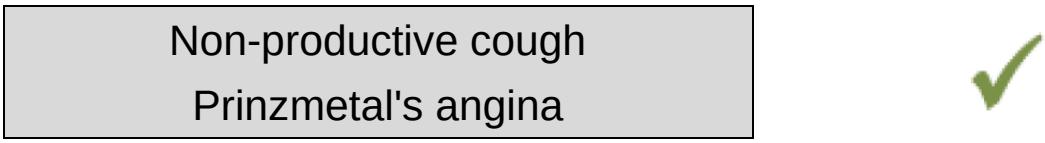


Figure 10: examples of SNOMED CT terms containing non-alphanumeric characters.

Rationale: as several non-alphanumeric characters are used commonly in every day noting the system should allow the user to continue to enter them.

SCT-022	Handle entered Abbreviations and Acronyms	Recommended Medium

The system should be able to handle the user entering abbreviation search terms (e.g. **aaa** for abdominal aortic aneurysm). It should not treat abbreviations as case sensitive (i.e. **TATT**, **Tatt** and **tatt** should be treated as equivalent entries).

Rationale: entering abbreviations or acronyms can provide a very quick and familiar way for clinicians to search for a term.

SCT-023	Provide progressive matching (performance permitting)	Recommended Medium

Progressive matching (where results are returned for each successive character that the user types in) should be provided where it does not cause any performance issues (i.e. react too slowly when done across a large SNOMED CT dataset).

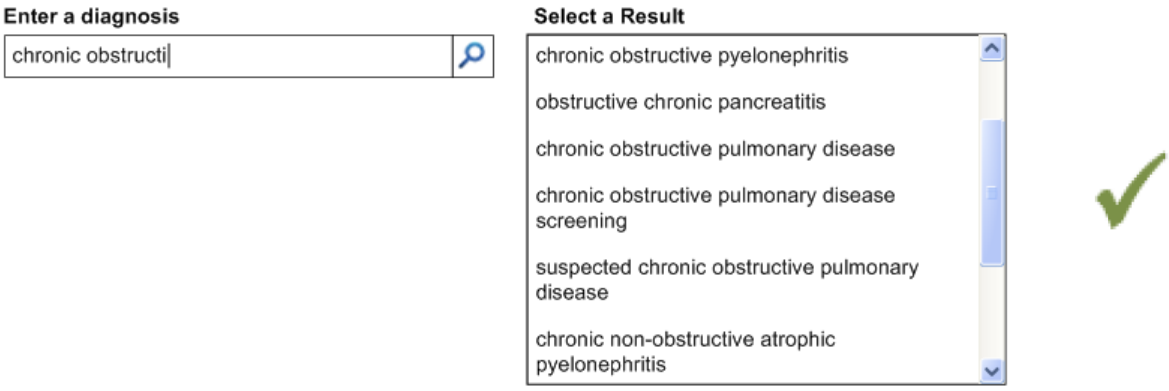


Figure 11: example of progressive matching functionality where results appear for every character entered in the text entry field.

© Crown Copyright 2011

Rationale: progressive matching can be very useful as it avoids the need for the user to type out their search term in full before seeing any results. It also allows them to be partly guided by the system and it can quickly alert the user to spelling mistakes made - with incorrectly spelt terms returning an immediate zero results.

SCT-024	**Do not offer auto-completion**	Recommended Low

Auto-completion offers suggestions to finish the words being typed in by the user. It can improve the user experience of searching by reducing the number of keystrokes a user has to make and to help reassure them that the system 'understands' their intentions.

Note the difference between auto-completion and progressive matching. Auto-completion completes a word that is only partially typed in. For example, a clinician starts to type in the search term **pneumonia**. After typing in a few characters, such as **pne**, the system offers a short list of words that it thinks will complete this word, such as **pneumonitis**, **pneumonia**, **pneumatic** and so on. This short list of words does not form part of the search results list, it is simply a list relating to the search text entry field. But, progressive matching does affect the search results list. For every character typed in, the search results list will update and show results matching what has been typed in so far.

Rationale: whilst auto-completion seems a useful idea, some early clinical CUI research has shown that it was distracting and could slow down the clinical noting process.

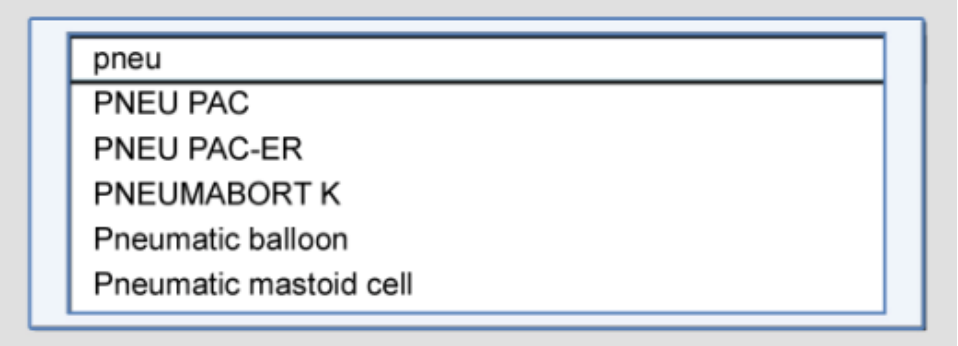


Figure 12: example of auto-completion when the search term **pneu** is entered

SCT-025	**Do not offer multiple search options**	Recommended Idea

In order to not overburden the user with many options and features, the search user interface should not offer other search options and choices to the user.

If additional search options are needed, then they should be offered as an advanced function which is not on direct display in the default user-interface, but is relegated to a less prominent position or on a separate screen.

In the example below, the user interface offers the user the ability to choose from 5 search mechanisms.

© Crown Copyright 2011

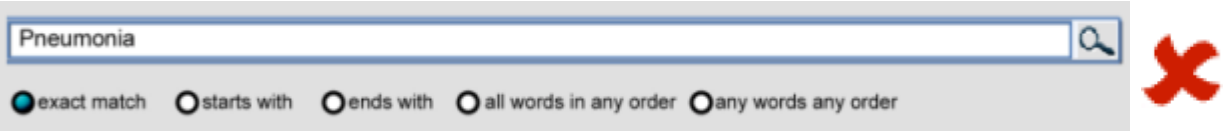


Figure 13: putting several search options on default display is not advised.

Rationale: whilst providing many ways of doing things may seem useful so that all situations can be accounted for, in most cases, less is more, as too much extra information on-screen takes more of the user's attention and slows them down.

Presenting several options also carries the risk that the user may change from the default search but then forget to change it back at a later moment. Also, that they change it and then cannot remember what their default setting was to set it back to.

SCT-026 Provide favourites	Recommended Medium

Alongside text entry, a useful function for users would be the ability to select SNOMED CT concepts from a favourites list. Favourites can provide a very useful, shortcut way of searching for SNOMED CT concepts.

Whilst the list of favourites tends to be those that the user has identified and chosen, other variations are also possible, such as:

- Terms chosen by the user
- Frequently used terms
- Recently used terms
- Local Department / Specialty terms
- Common terms (e.g. a set of common diagnoses)
- Conduit terms - a term that acts as a link to a collections of other terms (e.g. the term "eczema" could provide the user with a list of all SNOMED CT eczema terms)
- Hard-to-Find terms – a short crib-sheet of terms that - for whatever reason - are not easy to locate

Rationale: a favourites list can provide a quick and easy way for clinicians to select previously identified SNOMED CT concepts without having to perform a search for them.

© Crown Copyright 2011

3.3 Matching Concepts

This section presents suggestions relating to matching the entered search term with SNOMED CT concepts once a search has been initiated by the user.

SCT-027	**Where multiple search tokens are entered apply the AND operator**	Recommended Low

When multiple search tokens are entered (e.g. **prod cough**) the system should match terms that contain both tokens **prod** and **cough**. This means that terms containing only **prod** or only **cough** would not be returned as a result.

Rationale: the AND operator is proposed instead of the OR operator. This is because in early tests some users found it strange that when they entered in more search term text (e.g. 2 tokens instead of 1) that the number of results increased instead of decreased. They thought that by being more specific (providing the extra search token) that would refine their search and thus return fewer results as they were narrowing down the possibilities.

SCT-028	**Perform partial matching by default**	Mandatory Medium

The search string entered by the user should be handled as if it is a partial match (and not an exact match). For example, the system should see search string of **obes** as being **obes***.

Rationale: partial matching only requires the user to type in some of the characters of the wanted term – as opposed to exact matching that requires all characters to be typed in. And whilst exact matching does produce a much smaller, specific results list, any partial matching search can place exact matches to the top of the results list.

SCT-029	**Use STARTS WITH matching for each search token**	Recommended Low

The matching should look for terms that *start with* the entered search token. For example, when **pneu** is entered the system should search for terms starting with **pneu**. It should not look for terms that contain or end with **pneu**.

Similarly, if the search tokens **gast ref dis** are entered, then the system should only search for terms that have tokens starting with **gast**, **ref** and **dis**.

Rationale: it is assumed that most clinicians will prefer to search on prefixes rather than on word endings or words containing. Also, searching on words containing would produce a large number of returned results (e.g. searching on **ear** would return potentially many **heart** concepts).

© Crown Copyright 2011

SCT-030 Suppress frequently encountered tokens	Mandatory High

Each search performed will in general try to match the search tokens entered by the user with the tokens in each SNOMED CT concept held in the database. But, several of the tokens in each SNOMED CT concept in the database are so frequently encountered (such as **and**, **the**, **in** and so on), that when searched across they do not significantly help discriminate between a wanted and unwanted result.

Because of this, and in order to improve the search performance, these frequently encountered tokens should not be included in the search.

Note, that the user should still be able to enter in common words (e.g. **the**, **and**, **in** etc). This is because if the user types in **the** for example, then the system cannot know if the user is entering the common word **the**, or is entering the first few characters of the term **therapeutic**.

In order to provide a consistent search experience, the same list of frequently encountered tokens that are to be suppressed should be used by all software vendors.

Rationale: to improve the search performance common tokens should not be searched across.

SCT-031 Matching results	Mandatory Option

Option 1	**Return <u>all</u> lexically matching concept descriptions**

The matching results should be all SNOMED CT concept descriptions that lexically match the entered search term. For example, if **urinary infection** is entered then all concept descriptions containing tokens starting with **urinary** and **infection** will be returned as a matched result.

This means that several descriptions for a single concept could be returned in the results list significantly increasing the number of results returned.

acute sinusitis
acute inflammation of sinus
acute inflammation of nasal sinus
acute infection of sinus

Figure 14: four descriptions all from a single concept that could be returned by entering **acu sin**.

Rationale: users will be familiar with this search pattern - every result returned will include the search text they type in. Otherwise, users may find it strange to encounter results that do not contain the text they entered (e.g. the Fully Specified Name that does not include the search text, but is included in the results list because one of the synonyms in that concept does includes the search text). As it also provides all

© Crown Copyright 2011

descriptions from a concept, users can choose the SNOMED CT synonym that they prefer.

Option 2	**Return <u>one</u> item per unique concept where that concept has at least one lexically matching description**

In this option, one result per unique single concept is displayed. The result 'item' can then display one of the following:-

- Preferred term
- Synonym that is lexically closest to the search term (e.g. lowest Levenshtein distance from the search term)

The above items could also be prefixed with additional information, such as the SNOMED CT supercategory (taken from the Fully Specified Name) or the SNOMED CT chapter heading (taken from the transitive closure table).

Rationale: this matching method can potentially greatly reduce the size of the results list by not including all matching synonym descriptions.

Option 3	**Return <u>all</u> descriptions for a concept where that concept has at least one lexically matching description**

Where at least one description of a SNOMED CT concept lexically matches with the search term, then all the descriptions belonging to that concept should be included in the results list.

As shown in the example below, this means that for a single concept several descriptions – some of which won't lexically match the search term - could be returned in the results list.

atopic dermatitis
atopic eczema
allergic eczema
allergic dermatitis
disseminated neurodermatitis

Figure 15: four descriptions all from a single concept that could be returned by entering **atopic eczema**.

This option significantly increases the number of results returned. There are also issues relating to how any grouping or ordering of these descriptions is to be done, and whether the user need be made aware of that several descriptions exist for a single concept.

Rationale: whilst users may find it strange to encounter results that do not contain the search text they entered, this may be outweighed by the usefulness of having the choice of several different descriptions of a SNOMED CT concept.

© Crown Copyright 2011

SCT-032	Provide word equivalence matching	Recommended Medium

The system should by default (without any user prompting) perform word equivalence matching. By using equivalence tables, equivalent terms could be included in the results list. For example, if a clinician enters 'nose boil' then the equivalent term 'nasal furuncle' could also be provided in the results. This could be very useful for the user who may decide upon viewing the results that ‘nasal furuncle’ is the actual term they want.

As well as working for complete terms or phrases, the equivalence matching should also work for single input words. For example, the term 'rupture of cervix' could also return the equivalent 'tear of cervix' as the equivalence tables could recognise that the single words 'tear' and 'rupture' are equivalent.

Rationale: word equivalence matching allows clinicians to express phrases in multiple synonymous ways.

SCT-033	Highlight the use of word equivalence matching by the system	Recommended Option

Option 1	**Ensure user is aware that word equivalence matching is being done and allow them to control its usage**

When word equivalence matching is used, it may appear to users that it ‘breaks’ the normal matching applied. For example, a result may appear that does not resemble the entered search text (e.g. they enter **kidney failure** and see a result that does not include both these tokens, such as **renal failure**).which may cause confusion.

It then makes sense to inform users that word equivalence matching is being used. The system could:

- Clearly indicate where word equivalence matching is being used.
- Provide a control to allow the user to choose whether equivalence matching is switched on or off.
- Communicate which equivalent words the system is searching.

Rationale: so that confusion is avoided by users not thinking that the system is matching incorrectly or not as expected.

Option 2	**Do not advertise that word equivalence matching is being done nor offer any controls for it**

This option suggests not highlighting the fact that word equivalence matching is being used in the user interface. This means, not advertising that word equivalence matching is being used or not, nor providing any controls to allow the user to turn word equivalence matching on or off.

Rationale: this suggestion arises through a need not to clutter the user interface with additional controls and information which could lead to increasing the users cognitive load.

© Crown Copyright 2011

SCT-034 Inform the user that a search is taking place	Recommended Medium

The system should inform the user that a search is running. This is particularly important when searches take a long time to complete.

Rationale: the system should always keep users informed about what is going on through appropriate feedback. If a search runs in the background, without indicating this to the user, they user may think that the system is not working correctly or has crashed.

SCT-035 Provide a mechanism to cancel a running search	Recommended Medium

If a search takes a long time to complete then the system should provide a way for the user to cancel it mid-operation.

Rationale: users often choose system functions by mistake and will need to be able to back out of them easily. Any search that is started should be stoppable – especially if it takes a long time to complete.

© Crown Copyright 2011

3.4 Displaying Search Results

This section presents suggestions for displaying the search results.

SCT-036 Provide a scrollable results list	Recommended Idea

The search results should be able to be managed in a small list with scrolling. Vertically scrolling gives an easy and quick way to browse the results in a list. No horizontal scrolling should be necessary to see the information in the results table as the concept text should wrap onto a second line if necessary (see SCT-039).

Other user interface components are also available for displaying and browsing a list of results. Pagination (where the user has to click through individual pages of results) is another common design pattern. But, context depending, searching for a SNOMED CT concept using scrolling has the following benefits over pagination:

- Scrolling is generally better for allowing users to quickly scan results.
- The scrollbar control is standardized, so most users are already familiar with it. Pagination lacks standardization so it takes attention and learning to use it (e.g. locating where on-screen the paging buttons are).
- Scrolling allows users to move as little as one result at a time and they can scroll to anywhere in the list with a single interaction.
- When the scrolling results list is sorted, it is less likely to result in disorientation. With scrolling you can leave the scroll position where it is or you can automatically scroll to “chase” the list item that currently has focus; either way users can work with this.
- But, with pagination, users can be disoriented when sorting changes everything on their current page or if their page number changes to some other value to keep the current item in focus. Many paging implementations reset the user to Page 1 after sorting, which seems to avoid disorientation, but too often requires the user to re-navigate to a particular item.

Because of these pagination should only be used if absolutely necessary. Users are more likely to make their search more specific than use pagination.

Rationale: a small, vertically scrollable list provides a familiar control that allows users to quickly scan through a relatively large list of results. In the context of SNOMED CT encoding, such a list has several benefits over user interface alternatives.

SCT-037 Display at least 10 results simultaneously	Mandatory Medium

Sufficient on-screen room should be provided for the results list - it should be large enough to display at least 10 results simultaneously.

Rationale: this is to ensure that users are not searching through potentially a large list of results in a very confined space. This suggestion means that a sufficient

© Crown Copyright 2011

number of results can be displayed simultaneously – without forcing the user to perform a lot of scrolling.

SCT-038	Display a maximum of 20 results simultaneously	Recommended Idea

A maximum of 20 results should be visible on screen at once.

Rationale: as not to overwhelm users with too many results being displayed at once.

SCT-039	Results list width	Mandatory Medium

The results list width should allow at least 32 characters in a line. If the SNOMED CT concept exceeds this length, it should wrap onto a second line which is indented by two characters.

Rationale: this is taken from previous CUI guidance on presenting SNOMED CT concepts in a result list. The size of 32 characters allows the full text of about 50% of SNOMED CT labels (Preferred terms and Synonyms of active terms) to be displayed.

SCT-040	Displaying results longer than 2 lines	Mandatory Option

Testing has revealed that users prefer labels to be wrapped over two lines, rather than be truncated to fit a single line. This suggestion looks at the options for handling concepts that have a length greater than 64 characters and go beyond this second line (which is 15% of SNOMED CT concepts for Preferred terms and Synonyms of active terms).

Option 1	**Display the full SNOMED CT concept without truncation**

The concept should always be displayed in full – without truncation. This would mean that the concept should wrap to as many lines as it needs in the results list.

Wrapping to 3 lines would ensure that 94% of SNOMED CT concepts (Preferred terms and Synonyms of active terms) would be displayed in full.

Rationale: this option has the benefit that the entire term is immediately on display, without requiring the user to perform any other action to see its full text (e.g. hover mouse over term to get a pop-up tool tip). However, having several very long SNOMED CT descriptions could reduce greatly the number of concepts simultaneously visible in the results list.

Option 2	**Truncate the SNOMED CT concept**

If the concept text length exceeds two lines, the system must display all that it can over the two lines and then add an ellipsis (‘…’) at the end of the text to indicate that it has been truncated.

© Crown Copyright 2011

If this is done, the user should be given the ability to see the concept text in full by another mechanism, such as a tool-tip or pop-up display.

Rationale: previous CUI guidance references this option. It ensures that a results table can keep a fairly constant number of results showing simultaneously. However, this option carries the risk that with the full term text not on display, that the user chooses the wrong term. This risk could be mitigated by various means, including designing a user interface that highlights to the user when duplicate truncated descriptions are displayed.

SCT-041 Positioning the results list	Recommended Low

The results list should be located in an easily noticeable, consistent position. It should not obstruct any text being entered nor distract the user to such an extent that it obstructs the typing of further notes. Ideally, the results list should be displayed next to, or as close as possible to, the text input area (where the user entered their search term).

Rationale: having the text input area and results lists both visible and located close to each other not only provides a confirmation that the noting process is occurring, but also gives the user a reminder as to what search the displayed results are for.

SCT-042 Display the results as a flat list	Recommended Low

The primary, default method for displaying the concepts in the result list should be in the form of a ‘flat list’. This simply means a list that has no hierarchy.

Rationale: the results should not be hierarchically presented by default as there may be areas of the SNOMED CT hierarchy that do not fit with the user’s idea of relationships between concepts. This could result in users getting lost within the hierarchy.

For this reason, hierarchy browsing is not the primary method of displaying the results. The hierarchy browsing could be made available as an advanced or refining option for those instances where the user cannot find what they want, or are unsure about the precise labelling of what they want

SCT-043 Visually distinguishable concepts in results list	Mandatory High

Ensure that individual terms in the results list are clearly distinguishable from each other. This could be done by providing row highlighting or equivalent to make the individual results stand out from one another.

Rationale: to ensure that the user can easily and quickly scan through and select a concept.

© Crown Copyright 2011

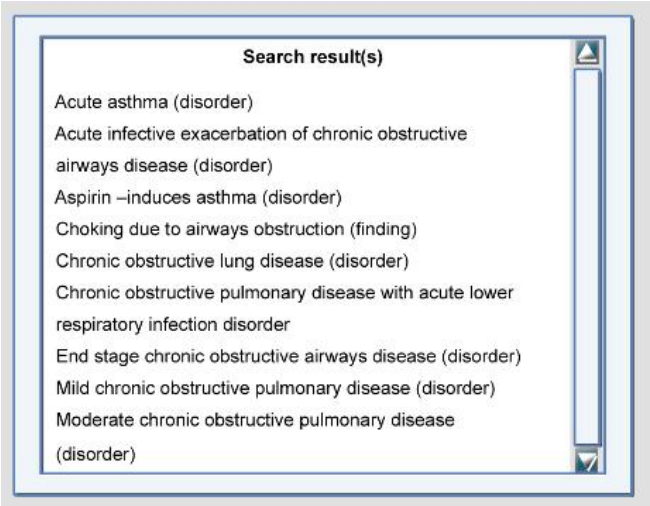


Figure 16: example showing terms not clearly appearing as separate entities.

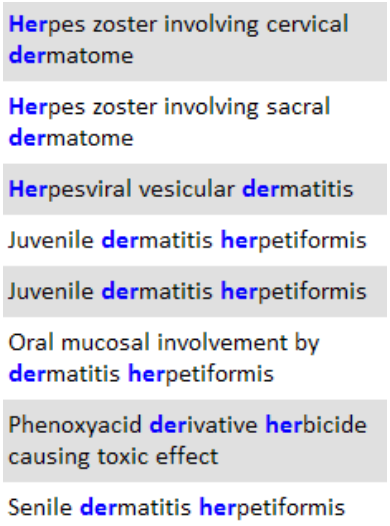


Figure 17: example using row highlighting to distinguish terms in the results list.

SCT-044	**Do not display SNOMED CT codes**	Mandatory Low

SNOMED CT concept or description Ids should not be displayed as part of the result in the user interface. **Rationale:** in most cases, there is no real need for users to know the codes behind their chosen terms. There exists also a small risk, that some users may choose a SNOMED CT concept by referring to its code – which carries a high risk of mis-selecting a term.

SCT-045	**Provide additional information per result**	Recommended Medium

The system should allow the user to view the Fully specified name for any concept. It should also communicate to the user where a result is a synonym.

Rationale: this additional information helps users to be able to distinguish between concepts when 2 different SNOMED CT concepts have the same description. Also,

© Crown Copyright 2011

stating when a result is a synonym is useful as, unlike the Preferred term description, the synonym label is often different to the Fully specified name label.

Atypical pneumonia
Atypical pneumonia



Figure 18: just showing two identical descriptions for two different concepts makes it difficult for the user to choose the wanted one.

Figure 19 displays two concepts with additional information contained within the result row of the table. This shows the Fully specified name, states when the concept is a synonym and also including the parent concepts.

Atypical pneumonia	Atypical pneumonia (disorder) *is an* Infective pneumonia
Atypical pneumonia	*synonym for* Lower primary atypical interstitial pneumonia (disorder) *is an* Interstitial pneumonia



Figure 19: example solution that provides additional information to help distinguish the 2 results.

Figure 20 shows an alternate way to displaying the additional information, in this case in a tool-tip.

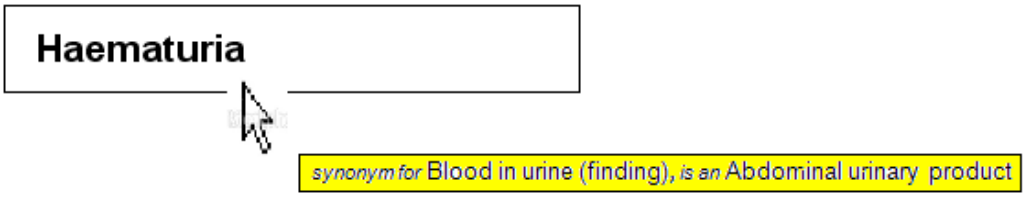




Figure 20: example with additional information in a tool-tip.

SCT-046	**Ensure all abbreviations and acronyms include an expanded text**	Recommended Medium

The UKTC editorial principles state that any abbreviation must also be accompanied by the expanded text. For example, figure 21 shows a description with abbreviation (**AAA**) and it's full, expanded text (**Abdominal aortic aneurysm**).

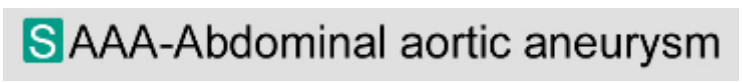




Figure 21: example of a description label with the abbreviation and its full text.

© Crown Copyright 2011

There are though, some allowed exceptions to this rule. Some abbreviations are deemed to be so unambiguous that they appear without any expanded text. An example of this is **IgA1** which appears in several concepts without the expanded text for **immunoglobulin A1**.

Outside of the allowed exceptions, there are a few cases where the abbreviated concept does not include a full text name as part of the description.

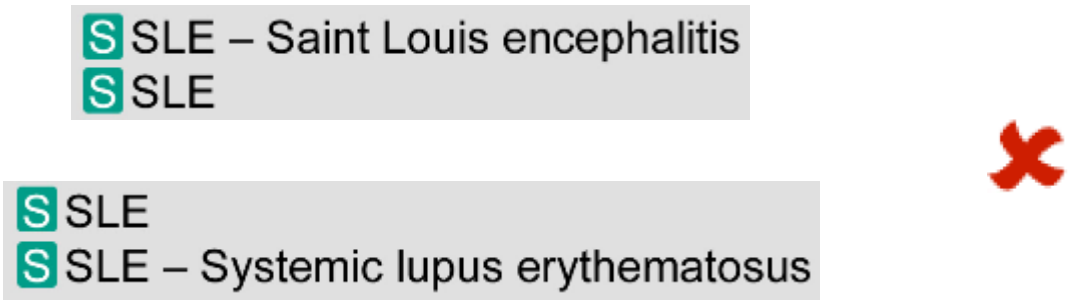


Figure 22: two examples where of two different concepts, having an acronym description of **SLE**, do not have a full text in their labels.

Users should not be presented concepts that use abbreviations without any expanded text as this introduces an obvious risk that the user could select the wrong concept by mistake.

Various solutions are available to remove this potential risk, ranging from the UKTC ensuring that no SNOMED CT concepts with abbreviations exist without an expanded text, or that the healthcare system appends the Fully specified name (minus the high-level category, such as 'disorder') to the existing description label.

Rationale: in order to reduce the risk that the user selects the wrong concept because multiple concepts share the same abbreviation or acronym, a full expanded text description should be included in the label.

SCT-047	**Highlight search tokens in the result**	Recommended Idea

For each result displayed, the part that matches with the search term should be highlighted.

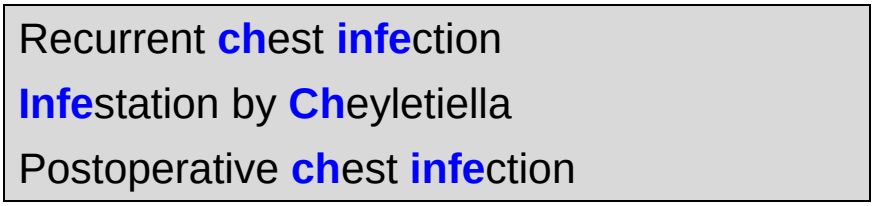


Figure 23: example showing search term **ch infe** highlighted in each description.

Rationale: this is to help users identify the tokens they searched for in the returned SNOMED CT descriptions.

© Crown Copyright 2011

SCT-048	Prioritise some concepts by putting them to the top of the results list	Recommended Idea

As some concepts are more likely to be chosen by the user, they should be given a higher priority than other concepts. This should be done by placing them at the top of the results list making them easier for users to locate. Some examples of these could be the following type of matches:-

- Exact matches
- Near matches
- Frequently used terms

There can be also different variations of 'Frequently used terms', such as being based upon frequency of terms chosen by the user, or by the local department, or frequency of use within a particular specialty.

Rationale: some concepts are more likely to be chosen by the user and so they should be given a higher priority to make them easier for users to locate.

SCT-049	Add appropriate headers within the results list	Recommended Idea

Where the results list has separate sections of results, headers should be used to highlight those sections. Figure 24 shows an example where the results list is made up of 2 parts: nearest matches and other matches. Here headers have been added to show this.

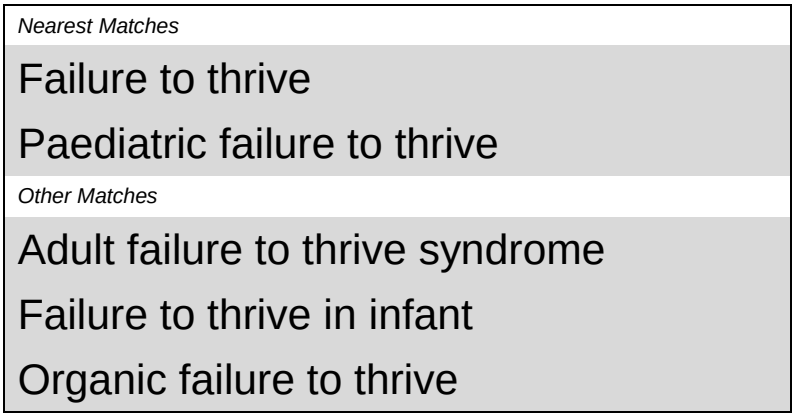


Figure 24: example of results list made up of 2 sections – nearest and other matches.

Rationale: so that users are aware when a results list is made up of multiple sub-results lists and can thus understand how the results are ordered.

SCT-050	Ordering the results list	Mandatory Option

Various strategies exist for ordering the results list. This suggestion presents the proposed options.

© Crown Copyright 2011

The ordering strategies outlined in this suggestion assume that any exact, near or frequently used matching results are given priority and appear at the top of the results list (as described in suggestion SCT-048).

Note, that whatever the ordering strategy used, it should not be one of the following:

- **Random ordering** – as this has no recognisable ordering pattern for the user to follow.
- **'Straight out of the database' ordering** – this is an ordering based upon the internal workings of the database containing the SNOMED CT data. Users won't recognise this ordering and in any case it may differ each time the data is accessed.
- **Ordering by SNOMED CT concept-id / description-id codes** – again this is not a recognisable ordering pattern.

Option 1	**Alphabetical**

The results list is alphabetically ordered.

Rationale: if the user can recognise what the ordering strategy is, it becomes easier for them to navigate through a large list of results. They know what to expect and can quickly skip through many results to get to the ones they want. An alphabetical ordering is such a strategy that is easy to recognise and use.

But, whilst an alphabetical ordering is very familiar, it does require that the user know what letter the SNOMED CT concept starts with, which is an obvious disadvantage.

Alphabetical ordering also does not take into account the search term typed in by the user - it gives no priority to concepts that are similar to what the user entered. For example, if the user enters **infant eczem** the concept **Infantile eczema** still only appears a third of the way down an alphabetical result list.

Option 2	**Lexical distance from search term**

Ordering based upon a measure of the mathematically calculated difference between the entered search term and the SNOMED CT term result.

Rationale: whilst this strategy is less familiar to most users, it does rank the results in relation to what the user entered as their search term.

So, whilst not being immediately easy to recognise – it does order based upon what the user has entered. Taking the previous example used, the term **Infantile eczema** would appear close to the top of the results list as it contains a similar number of characters to the entered search term **infant ecze**.

Option 3	**Number of characters**

Ordering is based upon the number of characters in the matched result. Where the number of characters is the same, alphabetical ordering could be used to further differentiate.

Rationale: less familiar ordering strategy, but again ordering is related to entered search text. Also, as many of the very long SNOMED CT concepts are typically not

© Crown Copyright 2011

that meaningful for recording typical day-to-day practices, this ordering strategy has the benefit that it suppresses them by ranking them lower down in the results list.

Option 4	**Number of words**

Ordering is based upon the number of words (tokens) in the matched result. Where the number of words is the same, alphabetical ordering could be used to further differentiate.

Rationale: again a less familiar ordering strategy, but where the ordering is related to entered search text. Also, as many of the very long SNOMED CT concepts are typically not that meaningful for recording typical day-to-day practices, this ordering strategy has the benefit that it suppresses them by ranking them lower down in the results list.

Option 5	**Frequency of use**

Ordering is based upon the frequency of usage of the concepts returned. This could be the frequency of usage for an individual, a department, an NHS Trust and so on.

Rationale: frequency of use ordering tends to work well if the wanted concept appears near to the top of the results list and is thus easily identifiable. If it does not, then they user has to browse through each item in the results list. Whilst doing this they cannot easily skip results or jump down the list as the ordering pattern will most likely be unfamiliar to them.

There is also an issue when many concepts are not frequently used – as they will all be ranked at the same level. Some additional ordering may be needed to further differentiate the results.

Figure 25 illustrates two examples of these ordering strategies.

SCT-051	**Provide ability to apply different ordering to results list**	Recommended Idea

In order to find the wanted concept, there may be a need for the user to be able to switch between ordering strategies. For example, it may be useful to switch between alphabetical to frequency of use ordering in certain situations.

If this suggestion is implemented, the user should not be offered too many different ordering strategies to choose from and it should always be clear in the user interface which ordering is currently being applied.

Rationale: providing the ability for the user to switch between ordering strategies could help locate difficult to find terms.

© Crown Copyright 2011

Option 1

Alphabetical

Nearest Matches
Dermatitis
Dermatomyofibroma
Dermatophytosis
Other Matches
Actinic **derm**atitis
Acute hand **derm**atitis
Adult atopic **derm**atitis
Allergic contact **derm**atitis
Atopic **derm**atitis
Atopic **derm**atitis of scalp
Chronic lip fold **derm**atitis
Congenital **derm**al sinus
Contact **derm**atitis
Contact **derm**atitis due to cyclohexane
Dermal cellular naevus
Dermal elastolysis
Dermatitis factitia
Diabetic **derm**opathy
Egg contact **derm**atitis
Generalised atopic **derm**atitis
Infantile restrictive **derm**opathy
Irritant contact **derm**atitis

Option 2

Lexical distance from search term

Nearest Matches
Dermatitis
Dermatophytosis
Dermatomyofibroma
Other Matches
Ring **derm**atitis
Rhus **derm**atitis
Sponge **derm**atitis
Atopic **derm**atitis
Senile **derm**atosis
Actinic **derm**atitis
Contact **derm**atitis
Dermal elastolysis
Diabetic **derm**opathy
Dermatitis factitia
Acute hand **derm**atitis
Egg contact **derm**atitis
Congenital **derm**al sinus
Dermal cellular naevus
Adult atopic **derm**atitis
Atopic **derm**atitis of scalp
Irritant napkin **derm**atitis
Chronic lip fold **derm**atitis

Figure 25: example of two ordering strategies for entered search term **derm**.

SCT-052	**Highlight favourites appearing in results list**	Recommended Idea

If the results list contains one of the user's favourites, this term should be highlighted so that the user is made aware of this.

Any highlighting should be done in such a way that the user's attention is drawn to this result, but without it being too distracting or adding too much visual complexity to the user interface.

Rationale: so that the user is made aware that a term they previously identified as a favourite is present in the results list.

© Crown Copyright 2011

SCT-053 Display number of results returned	Mandatory Medium

The system should display the number of results returned for each search.

Rationale: knowing the number of results returned (or number of pages of results returned) will affect how a user browses through those results. For example, if there are a small number of results returned the user may decide to browse through them all. But, if there is a large number returned, the user may decide to refine their entered search term and re-do the search.

SCT-054 Explicitly state when no results are found	Mandatory Medium

When no results are returned, the system should clearly state this – rather than simply displaying an empty results list. The system should also not encourage the user to enter an 'approximate' concept.

Rationale: the user should get useful feedback at all steps in the encoding process. This includes returning of results. If none are found, the results table should not simply be left empty – or worse still – left containing the previous search results. The system should clearly state what the outcome of each search has been.

SCT-055 With no results returned, offer further assistance	Recommended Medium

If no results are returned, the system should offer further assistance to the user. This can include where applicable the following type of item:

- Explain what the cause may have been for no results being found
- Describe possible actions the user can now take to improve the search
- Provide direct access or an easy way for user to take any presented actions

Rationale: rather than simply leave the user to guess as to why no results have been found, offer some insight along with possible actions the user can take to remedy the situation.

© Crown Copyright 2011

SCT-056 Number each search result	Recommended Idea

Each result in the results list should be numbered.

If this suggestion is implemented, the number displayed must look distinctly different to the SNOMED CT concept text. This is so that the user cannot confuse the number as being part of the SNOMED CT term and that the number does not distract from the scanability of the results list.

1. Make heart septal defect
2. Total hexosaminidase deficiency - adult
3. Total hexosaminidase deficiency - infantile
4. Fatty degeneration of heart
5. First degree burn of head
6. Second degree burn of face AND/OR head
7. Second degree burn of head



Figure 26: implementation showing numbered search results. Here the number could easily be mistaken as being part of the concept text and makes scanning harder to do.

11. B variant hexosaminidase A deficiency
12. B variant hexosaminidase A deficiency - adult
13. B variant hexosaminidase A deficiency - infantile
14. B variant hexosaminidase A deficiency - juvenile
15. B1 variant hexosaminidase A deficiency
16. Chronic viral hepatitis B without delta-agent



Figure 27: implementation showing numbered search results. Here the numbering looks distinctly different to the concept text.

Rationale: this is to provide an additional level of orientation, which is very useful if the user is browsing through a long list of results. Numbering would quickly allow them to recognise how far down the results list they currently are.

© Crown Copyright 2011

3.5 Refining the Search

In some situations, the user may not be able to find the exact term they are looking for. This could be because they cannot easily locate it in the results list or that it simply is not present in the list. When this occurs, it would useful to provide additional mechanisms that aid the user to find what they are looking for.

SCT-057	**Provide an option to expand beyond the default context**	Recommended Medium

In the case where the desired term cannot be found, users could be allowed to search outside of the default context (e.g. search beyond their current specialty across other specialties).

Rationale: for the situation where the sought after SNOMED CT concept does not exist in the current context, users should be allowed to search other contexts to help locate the concept.

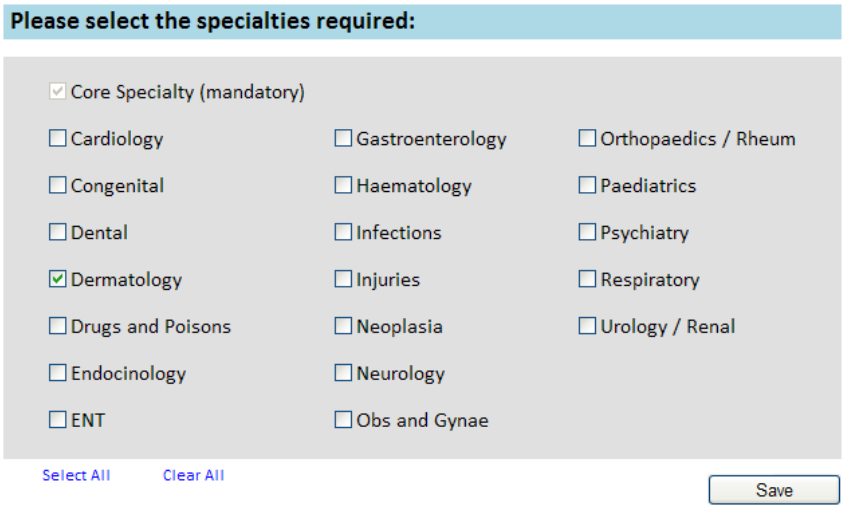


Figure 28: example of a system allowing the user to specify the specialties to take part in the search.

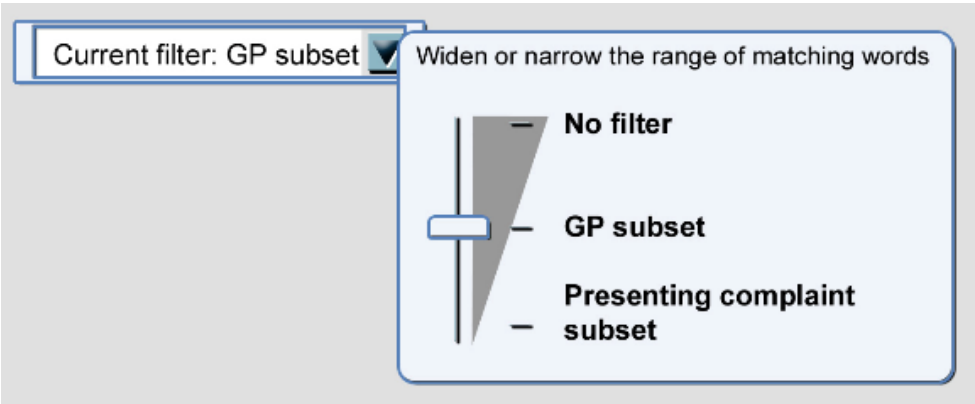


Figure 29: another example of a system that allows the user to select the extent of the search context using a filter slider.

© Crown Copyright 2011

Another alternative, is that the system automatically checks if any matches are available outside of the current context being searched. It could then inform the user of these matches.

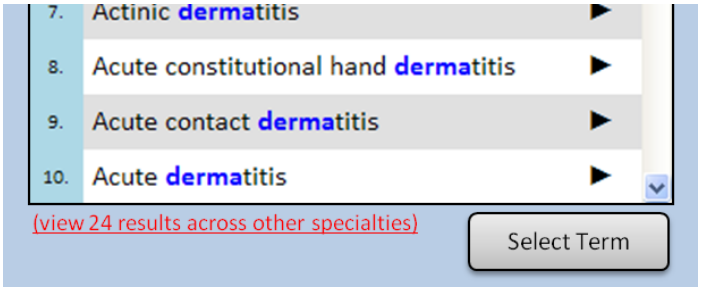


Figure 30: an example where the system has checked for matches across other specialties.

It is important that the system clearly communicates to the user what context is being currently used, so that they are aware of why the system is matching as it is.

SCT-058	**Provide hierarchical browsing for refinement**	Recommended Medium

Allow users to refine a concept in the results list by browsing other concepts that are hierarchically related to it. The related concepts could include the following:

- Parents of the selected concept.
- Children of the selected concept.
- Siblings of the selected concept (that is, the children of the concepts' parents).

The user interface design must clearly distinguish between the parents, siblings and children of a selected concept.

Rationale: this will allow users to select a more general or more specific instance of a concept. This can be useful when the user has located a concept which is not the one they want to encode, but is related to the one they want to encode. So, they can sue this concept to hierarchically browse to the term they want.

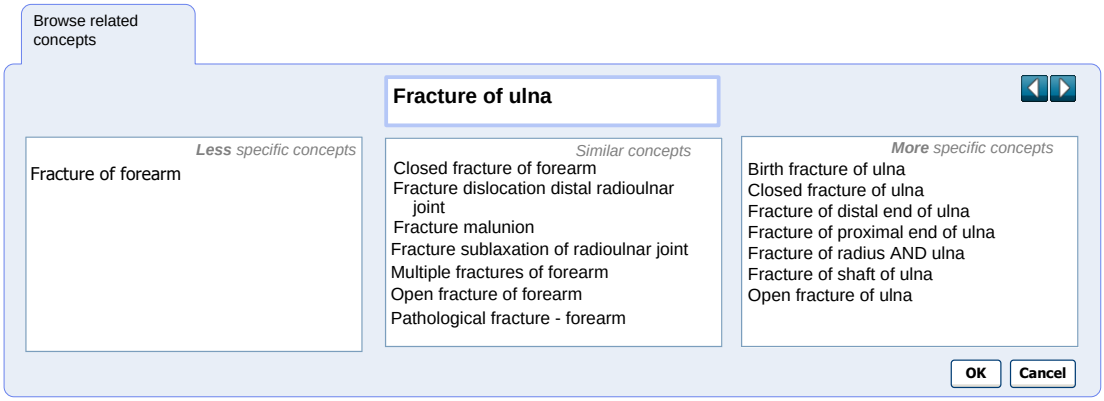


Figure 31: here the user can browse and choose from related concepts. Shown are parents (left panel), siblings (middle panel) and children (right panel) of the entered search term **Fracture of ulna**.

© Crown Copyright 2011

3.6 Selecting a Result

This section presents suggestions related to selecting a term from the results list.

SCT-059	**Allow control of the encoding process entirely by mouse, entirely by keyboard, and a combination both**	Mandatory High

As well as being controllable using a mouse, the encoding process should be controllable using the keyboard. The system should allow the user to trigger the search, navigate the results list and select a result using the keyboard. This should all be possible without requiring the user to firstly 'tab' or mouse-click to the necessary user interface component. This would include:

- Allow the <enter> key to trigger the search.
- When the result list is populated, the user should be able to directly give focus to and navigate up/down the results list.
- Allow the <enter> key to select a term from the results list and populate the search text field.

Rationale: users who are unable to, or prefer not to use a mouse need all functions to be available via the keyboard.

SCT-060	**Display the selected concept as the encoded concept**	Mandatory Medium

Once the user has selected a concept from the results list, the system will replace the user-typed 'original' search text with the SNOMED-CT concept.

Rationale: this is to prevent the confusion that can arise as in some systems, where the selected concept is different to the text finally displayed in the encoded field (e.g. the user selects a synonym from the results list, and the Fully specified name is entered as the encoded term). To prevent this, the term that the user selects from the results list should be displayed as the encoded term.

SCT-061	**Show in the text entry field that a concept has been encoded**	Recommended Idea

Users should be able to distinguish between text in the search text entry field that has been encoded, and text that has simply been typed into (which hasn't yet encoded).

This can be done by styling the text or the text entry field differently when it contains an encoded field or just normal text.

Rationale: so that it is immediately clear which fields the user has encoded.

© Crown Copyright 2011

SCT-062 Re-selecting a result	Mandatory Medium

Once a concept has been selected from the results list and this has been added to the search text entry field, the user should still be able to easily go back to the results list and select a different concept.

When the user does return to the results list, they should not have to re-enter their original search text. They should also not have to re-start the search. The results list should still present the results from the previously entered search text.

In addition, any other settings that the user had made (e.g. changed the default ordering, opted for hierarchical browsing, searching across other specialties and so on) should also be presented as they were previously.

Rationale: this functionality would useful if the user changed their mind, or selected the wrong concept by mistake. It also saves the user the time and effort of having to re-enter previously entered information.

© Crown Copyright 2011

4 Points for Further Discussion

This section notes some issues that arose during the creation of this document but were not (yet) added as suggestions. These points will need to be followed up on once further feedback has been received.

- **Local Shortcodes** – some clinicians will expect that a good user interface allows commonly used expressions to be entered using shortcuts. This would mean that local systems would need to create and manage their own list of shortcodes which are mapped to SNOMED CT concepts.

  Whilst this seems a handy option for users, it goes somewhat against the 'philosophy' of this document to provide a quick and easy matching without the need for clinicians to remember codes or shortcuts. More feedback is necessary on this topic.

- **Grouping of Results** – there is currently no suggestion for the grouping of concepts in the results list. Grouping by supercategory was considered, but omitted because it was assumed that a single context would be already set (SCT-005 Present only SNOMED CT concepts relevant to context) and that all results would only belong anyway to a single supercategory group.

  But, as suggestion exists for extending beyond the current context (SCT-057 Provide an option to expand beyond the default context) there may be a need to provide a grouping mechanism.

- **Switching Terminologies** – there may be legitimate exceptions to the suggestion SCT-002 (Do not require nor allow the user to choose the clinical terminology version). One such example could be in Mental Health where it may be necessary to switch between ICD-10 and SNOMED CT.

- **Technical definition** – for the suggestion SCT-003 (Exclude inappropriate content) it would be useful to provide a definition of which content should be excluded so that software vendors can easily prevent its inclusion.

- **Drag-and-Drop** – as a potential input mechanism alongside typing, cut & paste.

- **Highlight descriptions for a single concept** - if the results list contains more than one description of a single SNOMED CT concept there may be a benefit to identify to the user that those descriptions belong to the same concept.

- **Preferred term or Synonym** – whether there is any merit indicating to the clinician whether a results is a Preferred term or Synonym (for example, by prefixing the result with a **P** or an **S**).

- **Solutions** – to aid implementation, provide pragmatic solutions to those suggestions that are purely a statement of requirement.

- **Clinical Safety Group** – safety learning's from the Clinical Safety Group should be included in this document.

- **Expert user functionality** – some of the functions which are discounted (e.g. offering the user multiple search options), are discounted based on the assumption of a novice user. Depending on consultation feedback it may be necessary to look at more advanced functionality for expert users.

© Crown Copyright 2011

5 References

The following were referred to during the creation of this document.

1. CUI Design Guidance – Terminology – Display Standards for Coded Information, NHS Connecting for Health, Version 1.0.0.0 (July 2007).
2. CUI Design Guidance – Terminology – Elaboration, NHS Connecting for Health, Version 1.0.0.0 (July 2007).
3. CUI Design Guidance – Terminology – Matching, NHS Connecting for Health, Version 1.0.0.0 (July 2007).
4. Terminology Search and its effect on user acceptance, NHS Connecting for Health (Nov 2010).
5. Rules for truncation and wrapping of coded clinical data, Dr. Mike Bainbridge, Dr. Peter Johnson & Kit Lewis, NHS Connecting for Health (June 2008).
6. SNOMED Clinical Terms Technical Implementation Guide, the International Health Terminology Standards Development Organisation (Jan 2009).
7. GP Systems: Core Functionality Requirements (Document Record ID Key: NPFIT-FNT-TO-DSD-0052.03), NHS Connecting for Health.
8. User Interface Preferences in a Point-of-care Data System, Ronald B. Melles, M.D., Talmadge Cooper III, M.D., George Peredy, M.D.Permanente Clinical Information Systems, Kaiser Permanente Northern California Region (1998).
9. Getting Around in a Large Nomenclature File: Browsing SNOMED International, J. Craig Klimczak, D.V.M., Allen W. Hahn, D.V.M., Ph.D., MaryEllen Sievert, Ph.D. and Joyce A. Mitchell, Ph.D. (1994).

© Crown Copyright 2011

End of Document

© Crown Copyright 2011